

Brock Allan

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Education

University of California, Irvine

Bachelor of Science in **Computer Science**, GPA 3.6

- **Coursework:** Data Structures & Algorithms, Security & Privacy, Information Retrieval, System Design, Operating Systems, Computational Complexity, Differential Equations, Machine Learning
- **Distinctions:** Campuswide Honors Collegium, Published AI & Game Design Thesis, Dean's Honor List, CLAP certificate, Chancellors Excellence, FFA President, NHS President

Skills & Tools

- **Coding Languages:** Python, Java, JavaScript, C, C++, C#, Lisp, Prolog, HTML, Assembly, R
- **Industry Software:** Blender, Unity, Unreal Engine, Photoshop, Vim, Git, Spreadsheets, GDB

Technical Experience

Full-Stack Cybersecurity Lab & Network Intrusion Suite

March 2026 – June 2026

- Architected and led the implementation of an exploit and network analysis suite, leading a partner to identify **critical security vulnerabilities** across **software, web applications, and network layers**.
- Engineered a network intrusion detector in **Python** utilizing packet parsing libraries to process **multi-gigabyte .pcap** files, successfully flagging unauthorized **SYN port scanners** and **ARP spoofing** attempts from live traffic streams with a **98% success rate**.
- Designed and executed defenses against **control flow hijacking** exploits using **GDB**, that would otherwise bypass **stack-based memory protection** mechanisms such as **ASLR** and **DEP**, overriding return addresses and deploying **shellcode**.

Web Search Engine & Crawler

January 2026 – March 2026

- Architected and Lead development of a web search engine with **multi-threaded** crawler from scratch in **Python**, achieving **sub-300ms** query response times from a dataset of **50,000+ web pages**.
- Advised project members to engineer crawling logic that adheres to **Robots.txt politeness policies** and implemented **frontier-management** strategies to avoid **crawler traps** and **dead-end pages**.
- Spearheaded an optimized scraper to target and extract **high-density textual information**, creating a quality **inverted search index** under strict **operational and memory constraints**.

Custom Operating System

September 2025 – January 2026

- Developed a custom **Unix-like shell for x86/RISC-V in C** including a **recursive parser** that handles multi-stage **pipelines**, **I/O redirection**, and **comment-aware grammar** for automated script execution.
- Created **kernel level system calls** from scratch including **ps**, to monitor process states such as **Running**, **Sleeping**, and **Zombie**.
- Expanded **Exec** to identify **non-ELF text scripts** and automatically re-route execution through the designated **shell interpreter**.

Professional Experience

Learning Assistant, Intro to Programming, Shannon Alfaro, UCI

September 2025 - Current

- Instructed College students in how to code and design programs to achieve **foundational proficiency in core computing concepts**.
- Closely guided two groups of 6 students through technical coursework, hosted regular **office hours** to clarify complex topics and provided **detailed feedback** on assignments.
- Directed the LA team to **streamline communication**, provided **actionable feedback** to course instructors, and served as the **primary student advocate** to drive course improvements.

Computer Consultant & Data Analyst, Freedom Tax Insurance, Dana Burges

May 2020 – June 2024

- Oversaw the maintenance and repair of **mission-critical company infrastructure**, maximizing hardware lifecycle through **rigorous diagnostic testing**.
- Authored comprehensive executive reports, using **R**, that identified **operational bottlenecks** and delivered **data-backed recommendations** to senior leadership.